Întrebarea 537

Rezolvare

1. Metoda Fermat

$$\begin{array}{r|l} \sqrt{1.04.03} & 101 \\ \hline 1 & 2\underline{0}\times\underline{0}=0 \\ \overline{==4} & 20\underline{1}\times\underline{1}=201 \\ 0 & \\ \overline{403} & \\ 201 & \\ \overline{202} & \end{array}$$

Rezultă

$$[\sqrt{10403}] = 101$$

$$t = [\sqrt{10403}] + 1 = 102$$

$$t^2 - n = 102^2 - 10403 = 10404 - 10403 = 1 = 1^2$$

Deci

$$t = 102, \quad s = 1$$

$$b = t + s = 103$$

$$a = t - s = 101$$

Prin urmare

$$10403 = 101 \cdot 103$$

2. Quadratic Sieve

$$n = 10403$$

$$\begin{array}{r|l} \sqrt{1.04.03} & 101 \\ \hline 1 & 2\underline{0}\times\underline{0}=0 \\ \overline{==4} & 20\underline{1}\times\underline{1}=201 \\ 0 & \\ \overline{403} & \\ 201 & \\ \overline{202} & \end{array}$$

Deci $m = 101$

$$F(0) = 101^2 - 10403 = 10201 - 10403 = -202 = -2 \cdot 101$$

$$F(1) = 102^2 - 10403 = 10404 - 10403 = 1 = 1^2$$

Deci

$$102^2 \equiv 1^2 \pmod{10403}$$

$$102^2 - 1^2 \equiv 0 \pmod{10403}$$

$$(102 - 1)(102 + 1) \equiv 0 \pmod{10403}$$

$$101 \cdot 103 \equiv 0 \pmod{10403}$$

Calculăm $(101, 10403)$ și $(103, 10403)$

$$10403 = 101 \cdot 103 + 0, \text{ deci } (101, 10403) = 101$$

$$10403 = 103 \cdot 101 + 0, \text{ deci } (103, 10403) = 103$$

Rezultă că

$$10403 = 101 \cdot 103$$